

Efficient Encoding Scheme for Date Attachable Electronic Cash

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Abstract

Electronic cash (e-cash) plays an important role in the development of e-commerce. Not only can the e-cash preserve the advantages of traditional paper cash, like the untraceability property, but it has some characteristics lack in the traditional one, such as the convenience property, as well. In this manuscript, we propose a new date encoding method for e-cash without breaking any essential property of the e-cash. The proposed date-attachable e-cash scheme allows a payer to attach the desired date to an e-cash during a transaction, and with the aid of the attached date, the payee can charge the interest from the bank more easily. Furthermore, the proposed scheme satisfies the attachability property, and the attached date will not be modified by anyone else in any malicious way. Besides, in order to achieve flexibility, the proposed scheme is independent of the underlying partially blind signature technique, and hence it can be implemented easily by using any existing partially blind signature scheme as long as it is secure.

Keywords: Blind Signatures, Electronic Cash, Security & Privacy, Cryptography

1 Introduction

As the past generation goes by, the computer techniques step into a new generation simultaneously. With the aid of the advanced computer and network technologies, the e-commerce has a great growth. Not only do more and more customers accept the electronic transactions but they are also willing to adopt the advanced techniques. Among the transactions in e-commerce, the electronic cash plays a major role, which can be used as a medium of the transactions, while the payer and the payee are in distributed status.

Electronic cash was introduced by Chaum [3]. The basic architecture of an electronic cash scheme contains three kinds of parties, a group of payers, a group of payees, and a bank. Besides, there are three basic stages in the e-cash protocol,

initialization, withdrawing, and payment. In the initialization stage, the bank will prepare its own public-private key pair, and publish the public key along with some related information. In the withdrawing stage, a customer withdraws the digital money, which is blinded, from her/his account in the bank, and then the customer performs the unblinding phase to obtain the valid one. Finally, when the customer performs a transaction with a shop, she/he transmits the e-cash to the shop. After receiving the e-cash, the shop will deposit the e-cash in the bank by forwarding the e-cash to the bank. The bank will verify the correctness and perform the double-spending checking on the e-cash after receiving it. Once the verification process is finished successfully, the bank will inform the shop that the transaction is valid, and then the shop will inform the customer that the payment is successful.

Since the e-cash(s) are a sequence of digital signals, they are easily duplicated. Therefore, the bank makes use of double-spending checking to ensure that every e-cash in any transaction is fresh without being spent twice or more. However, in order to carry out the double-spending check, the bank has to record every accepted e-cash in its database. It will result in unlimited growth of the bank's database. In order to cope with such a problem, the partially blind signature technique is developed, where an expiration date is attached into every issued e-cash by the bank. The bank can remove the e-cash from its database if the current date is greater than the expiration date attached on the e-cash,

Several related works about e-cash have been presented [1][4], but rare of them focus on date attachable e-cash, which makes it possible for the payer to attach a desire date on the e-cash during a transaction. The dynamically attached date can be set as the depositing date of the e-cash to help the owner with calculating the interest she/he deserves. Besides, in order to prevent any loss of the payer and the payee, the attached date cannot be modified by any malicious party to guarantee the attachability property [6].

In this paper, we propose an encoding method to encode the attached date of an e-cash. Not

only is the proposed scheme effective to attach a desire date on the e-cash by the owner, but also the e-cash protocol still satisfies all essential properties, such as the unlinkability, the unforgeability, and so on. Moreover, the proposed idea is independent of the underlying signature fundamental primitives, i.e., any existing e-cash scheme based on partially blind signatures can be adopted to build the protocol as long as it is secure.

The rest of the paper is organized as follows. In Section 2, we review a partially blind signature scheme, which will be taken as the basic foundation of our scheme. In Section 3, the proposed protocol is demonstrated. In Section 4, the properties and performance analysis of the proposed scheme will be presented. Finally, a concluding remark is given in Section 5.

2 Generic Partially Blind Signature Scheme

In this section, we define a generic partially blind signature scheme based on Fan-Chen generic blind signature scheme [7]. It will form the underlying signature foundation of the proposed date-attachable e-cash scheme.

A partially blind signature scheme basically contains two kinds of participants, a signer and a group of users who want to obtain the signature from the signer. Different from the typical blind signature scheme, the signer and all users have to negotiate and agree on a set of predefined-format information in advance. And the signer must ensure that every issued signature contains the predefined information without being modified by anyone else. The protocol is described below.

Let M be the underlying set of messages, R be a finite set of random strings, and W be finite set of strings with the predefined format which is negotiated by the signer and all users in advance. A generic partially blind signature scheme consists of five elements (B, H, S, U, V), and they are described as follows:

1. $H: M \rightarrow M$ is a public one-way hash function. Given y , it is computationally infeasible to derive m such that $H(m) = y$, and it is also computationally infeasible to find two distinct messages m_1 and m_2 such that $H(m_1) = H(m_2)$.
2. $S: M \times W \rightarrow M^K$ is a signing function which should be kept secret by the signer where K is a positive integer. Given a message $m \in M$ and a predefined-format information $w \in W$, it is computationally infeasible to form $S(H(m), w)$ or modify m

and w embedded in $S(H(m), w)$ without S , where $S(H(m), w)$ is called the signer's signature on message m and the predefined-format information w .

3. $V: M^K \times M \times W \rightarrow \{\text{true}, \text{false}\}$ is the public verification formula. $V(t, m, w) = \text{true}$ if and only if t is the signer's signature on m with the predefined-format information w . Hence, $V(S(H(m), w), m, w)$ is always true for each $m \in M$ and $w \in W$.
4. $B: M \times R \rightarrow M$ is the blinding function. It is impossible to find out the message m from $B(m, r)$ without r , where $r \in R$ is randomly chosen and kept secret by the user who wants to request a signature. The string r is called the blinding factor of m and $B(m, r)$ is said to be the blinded message.
5. $U: M^K \times R \rightarrow M^K$ is the unblinding function. For each $m \in M, r \in R$ and $w \in W$, $U(S(B(H(m), r), w), r) = S(H(m), w)$. It is impossible to decide $S(H(m), w)$ from $S(B(H(m), r), w)$ without r .

The details of the partially blind signature protocol are described as follows:

1. **Blinding:** First, a user chooses a message $m \in M$, randomly selects a blinding factor $r \in R$, and prepares a string $w \in W$. Then the user computes $\alpha = B(H(m), r)$ and submits the blinded message (α, w) to the signer.
2. **Signing:** After receiving (α, w) , the signer verifies that w is in W , and then computes $t = S(\alpha, w)$, and sends t back to the user. The parameter t is called the partially blind signature because the message digest $H(m)$ embedded in α is unknown to the signer but w is clear from the signer's point of view.
3. **Unblinding:** After receiving t , the user derives $s = U(t, r)$ which is equivalent to $S(H(m), w)$.
4. **Verifying:** The parameter s is the signer's signature on message m with the predefined-format information w . The triple (s, m, w) can be verified by checking whether the verification formula $V(s, m, w)$ is true or not.

The generic partially blind signature scheme has two important features:

1. Nobody can produce this triple (s, m, w) without the signing function S . This is the *unforgeability* property.
2. It is information-theoretically impossible for the signer to derive the link between a signature s and the instance of the signing protocol which produces the partially blinded form, $S(B(H(m), r), w)$, of s without r under the same predefined information w . In other words, under the identical w , all of the signatures s 's are indistinguishable from the signer's point of view as long as the blinding factors r 's are kept secret by the users. This is referred to as the *unlinkability* property under the same predefined information.

3 The Proposed Encoding Method For Date Attachment

In the proposed encoding method, we make use of the expiration date, attached to the partially blind signature, to help us to correctly indicate the correct date of the transaction. With the aid of the attached expiration date and the interval between the expiration date and transaction date, everyone is able to calculate the actual transaction date. Besides, this scheme is efficient and effective against some previous drawbacks, such as the collusion problem, the Y2K problem, and incorrect interest calculation. The related discussions will be given in Section 4.

In Section 3.1, we present our scheme based on the generic partially blind signature shown in Section 2. In addition, we adopt Abe's partially blind signature scheme as the underlying signature foundation to further demonstrate our scheme in Section 3.2.

3.1 The Generic Version of The Proposed Scheme

The proposed scheme contains four stages, i.e., initializing, withdrawing, unblinding, and depositing. The details are described as follows.

1. **Initializing.** The bank makes the formats of the functions $\{B, H, U, V\}$ of the generic partially blind signature public and keeps the signing function S secret. In addition, let each issued e-cash be worth y dollars.
2. **Withdrawing.** A payer randomly selects a number r as the blinding factor and chooses two strings x_i , $1 \leq i \leq 2$, where r and all x_i 's must be kept secret by the payer. Then, the payer computes

$$\begin{cases} m = H^b(x_1) || H^b(x_2) \\ \alpha = B(H(m), r) \end{cases}$$

and sends α to the bank where we assume that b is more than the lifespan of every issued e-cash, i.e., $b > (\text{lifespan} + 1)$. For example, if the lifespan of each e-cash is 366 days, b is at least 368. Then the bank signs on α along with the expiration date w to form the signature $t = S(\alpha, w)$, and returns t back to the payer. Finally, the bank deducts y dollars from the payer's account.

3. **Unblinding.** After receiving t , the payer performs the unblinding operation by computing $s = U(t, r)$, and then the payer obtains the triple (s, m, w) , which is an e-cash worth y dollars.

4. **Depositing.** While the payer decides to pay the e-cash to the payee during some transaction, the payer attaches the difference of days, a , between the current date and the expiration date to the e-cash by computing $c_1 = H^{a+1}(x_1)$ and $c_2 = H^{b-(a+1)}(x_2)$, where the current date plus a days equals the expiration date embedded in w . For example, if the current date is March 15, 2007 and the expiration date in w is March 14, 2008, then $a = 365$. Then the payer sends the date attached e-cash (a, s, c, w) to the payee where $c = \{c_1, c_2\}$. The payee can verify the correctness of the e-cash and the attached date by checking whether the formula

$$V(s, H^{b-(a+1)}(c_1) || H^{(a+1)}(c_2), w) = \text{true}$$

holds or not. If the e-cash is correct, the payee forwards the e-cash (a, s, c, w) to the bank for double-spending checking. The bank checks that

- The e-cash is not expired according to the expiration date w embedded in the e-cash;
- $V(s, H^{b-(a+1)}(c_1) || H^{(a+1)}(c_2), w) = \text{true}$; and
- the current date plus a days equals the expiration date w .

Finally, if the e-cash is fresh and the above three conditions are true, the bank will store the date-attached e-cash in its database and add y dollars to the payee's account and then notify the payee that the transaction is successful.

3.2 The Proposed Scheme Based on Abe's Partially Blind Signature

The four stages are described below.

1. **Initializing.** Let the length of the predefined information be $k-2$ bits. The bank selects two large distinct primes p, q , and then computes $n = pq$, where $s_i \nmid u$ for all primes ($3 \leq s_i \leq 2^k-1$) where $u = \text{lcm}((p-1), (q-1))$. Then the bank selects a prime e with $2^{k-1} \leq e$ as the public key and its corresponding private key d satisfies $ed = 1 \pmod u$ [1]. Thus, (e, n) and (d, p, q) are the bank's public and private keys, respectively. In addition, a one-way hash function H is published along with the public key (e, n) . Finally, let each issued e-cash be worth y dollars.

2. **Withdrawing.** A payer randomly selects an integer $r \in Z_n^*$ as a blinding factor, and then randomly chooses two strings x_i , $1 \leq i \leq 2$ where all x_i 's and r are kept secret. The payer computes

$$\begin{cases} m = H^b(x_1) || H^b(x_2) \\ \alpha = r^{eF(w)} H(m) \pmod n \end{cases}$$

and sends (α, w) to the bank, where w is the expiration date and F is a formatting function with $\gcd(F(w), u) = 1$ [1]. Furthermore, we assume that the lifespan of every issued e-cash is less than b days. After receiving α , the bank computes $d_w = (eF(w))^{-1} \pmod u$ and signs on α to obtain $t = \alpha^{d_w} \pmod n$ and returns t back to the payer. Then the bank deducts y dollars from the payer's account.

3. **Unblinding.** After receiving the signing result t , the payer computes $s = r^{-1}t \pmod n$ to obtain a signature triple (m, s, w) , where (m, s, w) is an e-cash worth y dollars.

4. **Depositing.** While the payer decides to pay to the payee during a transaction, the payer attaches the current date to the e-cash by computing $c_1 = H^{a+1}(x_1)$ and $c_2 = H^{b-(a+1)}(x_2)$, where the definition of a is identical to that described in Section 3.1. Then the payer sends the date attached e-cash (a, s, c, w) to the payee, where $c = \{c_1, c_2\}$. The payee can verify the correctness and the attached date of the e-cash by checking the formula

$$s^{eF(w)} = H^{b-(a+1)}(c_1) || H^{a+1}(c_2) \pmod n.$$

If the e-cash is correct, the payee forwards

the e-cash (a, s, c, w) to the bank for double-spending checking. The bank also checks that

- The e-cash is not expired according to the expiration date w embedded in the e-cash;
- $s^{eF(w)} = H(H^{b-(a+1)}(c_1) || H^{a+1}(c_2)) \pmod n$; and
- the current date plus a days equals the expiration date w .

Finally, if the e-cash is fresh and the above three conditions are true, the bank will store the date-attachable e-cash in its database and add y dollars to the payee's account and then notify the payee that the transaction is successful.

4 Discussions

In this section, we will give some analysis on the characteristics and performance of the proposed date-attachable e-cash scheme. Table 1 illustrates the comparison result between the proposed scheme and other related schemes [2][5][6][8].

- **How to calculate the deserved interest for the owner of an e-cash after depositing.** In a transaction, the payer transmits the e-cash (a, s, c, w) to the payee, so that the owner of this e-cash will be changed from the payer to the payee. After the transaction, the payee will deposit this e-cash in her/his account. From this moment, the bank will start to accumulate the interest of the e-cash for the payee according to the attached date of the deposited e-cash. As to the details of interest calculation (such as the interest rate, the calculation formula, and so on), they are beyond the scope of this paper.
- **The correctness of attached dates and interest calculation.** During a transaction, a payer will attach the current date on his e-cash and form the 4-tuple (a, s, c, w) , where a stands for the number of days between the current date and the expiration date w . Everyone has the ability to calculate the attached date as the example presented in Section 3.1. Therefore, if the payee intends to conspire with the payer to cheat the bank by attaching another date, the payment will be declined immediately by the bank since the current date plus a does not equal the expiration date w . Furthermore,

in our scheme, the date is attached as the form of $m = H^{b-(a+1)}(c_1) || H^{(a+1)}(c_2)$, and hence anyone else including the bank cannot modify a without getting the strings $\{x_1, x_2\}$, which are randomly chosen by the payer. Thus, the interest can be calculated correctly according to the correct attached date, and neither the payee nor the bank will suffer from any loss.

- **Free from the Y2K problem.** In our scheme, the attached date is represented by the interval between the current date and the expiration date w . Therefore, it is free from the Y2K problem.
- **Communication cost of the payment.** In the proposed scheme described in Section 3.2, during the payment process, the payer sends the date-attached e-cash (a, s, c, w) to the payee, where the length of the date interval a is usually less than 9 bits and under SHA-256, $|c| = |c_1| + |c_2| = (256 + 256) = 512$ bits. Consequently, under a 1024-bit modulus n , the size of (a, s, c, w) equals to $|a| + |c| + |s| + |w| = 1575$ bits $= 1.54|n|$, since the length of the expiration date w is usually less than 30 bits. The comparison of the communication cost between ours and other related schemes is summarized in Table 1.

5 Conclusion

In this paper, we have designed an encoding method based on the expiration date to help the payer to attach the desired date on an e-cash during a transaction. Not only can the scheme guarantee the correct interest deserved by the cash owner, but also it is free from the problems of previous schemes, such as the Y2K problem, and the heavy-load computation and communication cost. Furthermore, we adopt a generic presentation to illustrate the proposed scheme, which is independent of the underlying signature foundations. In other words, the proposed date-attachable e-cash protocol is with high

flexibility since it can be implemented by any partially blind signature scheme as long as it is secure.

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References

- [1] M. Abe and E. Fujisaki, "How to date blind signatures," *Advances in Cryptology-ASIACRYPT'96* LNCS, 1163, Springer-Verlag, pp. 244-251, 1996.
- [2] C. C. Chang and Y. P. Lai, "A flexible date-attachment scheme on e-cash," *Computers & Security*, vol. 22, no. 2, pp. 160-166, 2003.
- [3] D. Chaum, "Blind signatures for untraceable payments," *Advances in Cryptology-CRYPTO'82*, Springer-Verlag, pp.199-203, 1983.
- [4] D. Chaum, A. Fiat, and M. Naor, "Untraceable electronic cash," *Advances in Cryptology-CRYPTO'88*, LNCS 403, Springer-Verlag, pp.319-327, 1990.
- [5] W. K. Chen and C. I. Fan, "Blind one-time signature and its applications," *Journal of Discrete Mathematical Sciences & Cryptography*, vol. 7, no. 1, pp. 55-69, 2004.
- [6] C. I. Fan, W. K. Chen and Y. S. Yeh, "Date attachable electronic cash," *Computer Communications*, vol. 23, pp. 425-428, 2000.
- [7] C. I. Fan and W. K. Chen, "An efficient blind signature scheme for information hiding," *International Journal of Electronic Commerce*, vol. 6, no. 1, pp. 93-100, 2001.
- [8] W. S. Juang, "D-Cash: A Flexible Pre-paid E-cash Scheme for Date-attachment," *Electronic Commerce Research and Applications*, in press, New York, Elsevier Press, 2007.

Table 1: The comparisons among [2], [5], [6], [8], and the proposed scheme

	Chang-Lai [2]	Chen-Fan [5]	Fan-Chen-Yeh [6]	Juang [8]	Ours
P_1	Yes	Yes	Yes	No	Yes
P_2	No	Yes	Yes	Yes	Yes
P_3	Yes	No	No	Yes	Yes
P_4	$\approx 2.26 n ^*$	$\approx 2.01 n $	$\approx 2.51 n $	$\approx 2.06 n $	$\approx 1.54 n $
P_1: Independent of the underlying signature schemes P_2: Correct attached dates and interest calculation P_3: Free from the Y2K problem P_4: The total communication cost of the payment *n denotes the bit length of the modulus n.					